

Open-NL: A Transparent Computational Testbed for Wide Dynamic Range Compression

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Abstract

Open-NL is an open-source computational testbed designed for the transparent modeling, component-level ablation, and evaluation of Wide Dynamic Range Compression (WDRC) prescriptive rules. While modern prescriptions like NAL-NL2 rely on extensive empirical regularizations to ensure clinical safety—historically necessitated by the well-documented failures of theoretical intelligibility maximization—their compiled, closed-source nature prevents researchers from isolating the interactions of specific heuristic safeguards. Open-NL addresses this limitation by coupling a multi-start Nelder-Mead effective (desensitized) Speech Intelligibility Index (SII) optimizer with the Moore & Glasberg (2004) specific-loudness model within an established 7-profile evaluation paradigm. We benchmark this testbed across a $4^4 = 256$ -permutation heuristic sweep alongside a 5-iteration multi-start stability analysis, establishing a fundamental distinction between numerical convergence and heuristic sensitivity: while the multi-start solver exhibits high numerical stability ($SD < 0.025$ SII) for any fixed parameter set, prescribed targets display extreme heuristic sensitivity in severe-sloping

configurations, swinging modeled monaural loudness between 0.9 and 5.3 sones under minor rule modifications. Because standard monaural models systematically underestimate real-world binaural broadband summation in impaired populations, this modeled loudness frontier is strictly illustrative rather than clinically definitive. When stripped of clinical heuristics, the optimizer’s raw mathematical drive to maximize desensitized SII dramatically inflates high-frequency gain in profound profiles, demonstrating why the SII metrics generated by such optimizers must be interpreted strictly as internal convergence diagnostics rather than as comparative clinical endpoints. By exposing these trade-offs within an inspectable framework, Open-NL provides the computational substrate for future stochastic calibration workflows designed to replace static heuristic boundaries with individualized, distortion-aware objective functions.

I. INTRODUCTION

Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice. While earlier investigations suggested that generic targets might outperform proprietary first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contemporary evidence indicates that aided speech recognition in noise often shows no significant difference across formulas. However, while formula choice has relatively modest intelligibility consequences in background noise, it drives substantial variations in overall loudness, making modeled loudness (quantified in sones per the Moore & Glasberg 2004 impaired loudness model) the primary dependent variable in prescriptive evaluation.

While the derivations and optimization objectives of major algorithms like NAL-NL2

36 and DSL m[i/o] are published in detail, their compiled software implementations remain
 37 closed-source. Audiological science has long recognized that unconstrained intelligibility max-
 38 imization fails clinically without extensive empirical regularization. The historical evolution
 39 from NAL-NL1 to NAL-NL2 specifically required empirical corrections—including global
 40 gain reductions (-2 dB for female listeners and new users) and reduced compression ratios for
 41 severe losses—to counteract aggressive over-amplification provoked by pure mathematical
 42 optimization (Keidser, Dillon, Carter, & O’Brien, 2012a). However, because clinical fitting
 43 software packages are compiled black boxes, researchers cannot isolate the relative contri-
 44 bution of individual heuristic rules or observe how specific safeguards interact within the
 45 optimization cascade. Existing open-source tools serve distinct functional niches: openMHA
 46 (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather
 47 than a target generator; the Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010)
 48 provide rigidly defined equation-based targets rather than a modular optimization sandbox;
 49 and the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers loudness modeling
 50 without native prescriptive inversion. Consequently, investigators cannot isolate a specific
 51 prescriptive heuristic within an optimization loop without reverse-engineering an entire
 52 proprietary engine. Open-NL fills this gap by providing a modifiable R substrate explicitly
 53 designed for the modular ablation of prescriptive heuristics. By coupling a Nelder-Mead
 54 desensitized SII optimizer to an integrated C++ specific-loudness engine, researchers can
 55 systematically disable, isolate, or invert individual heuristics—such as evaluating the upward
 56 spread of masking when disabling the 30 dB conductive safety cap. This modular architecture
 57 aligns directly with evolving audiological frameworks, such as the multi-profile philosophy
 58 introduced in NAL-NL3 (Kitterick et al., 2026).

Crucially, to benchmark this testbed without introducing confounding variables, the optimization layer is embedded within a strictly reproduced evaluation paradigm. The seven reference audiometric profiles, the Moore & Glasberg (2004) specific-loudness model, and the ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework established by Johnson and Dillon (2011). (Throughout this manuscript, “ANSI SII” refers to raw physical audibility, whereas “desensitized SII” or “desensitized SII” refers to audibility incorporating severe-loss desensitization and level distortion penalties). Because this physiological evaluation space is already established in the literature, the primary contribution of this manuscript is the transparent computational testbed itself. By exposing the behavior of numerical solvers within this standardized sandbox, we clarify a crucial distinction: while numerical solvers converge stably on any fixed objective space, theoretical WDRC target generation exhibits acute parameter sensitivity to uncalibrated heuristic boundaries, providing the computational infrastructure necessary to quantify and calibrate these interactions.

Clinical Safety and Usage Disclaimer

It is imperative to state unambiguously that Open-NL is strictly a computational research testbed and **must not be used for fitting hearing aids on human listeners in its current form**. Because the framework deliberately permits aggressive, over-prescriptive targets for boundary testing—such as utilizing an aggressive 60 dB HL severe-loss booster onset that permits 14–15 sones of theoretical loudness for profound losses—it carries a significant risk of severe over-amplification. As established by Ching, Dillon, Katsch, and Byrne (2001), aggressive high-level targets in steeply sloping or profound losses are precisely where over-amplification risks are greatest, as the effectiveness of high-frequency audibility

severely degrades as hearing loss worsens (desensitization). The hypotheses and targets generated by Open-NL represent extreme mathematical boundaries intended to trigger experimental loudness rejection in controlled research settings, not clinical solutions. Any future behavioral translation of this framework requires independent institutional review, with mandatory real-ear verification and strict, individualized loudness-tolerance limits implemented as absolute prerequisites.

II. ALGORITHM ARCHITECTURE

A. Prescriptive Rationale and Objective Function

The choice of objective function represents the primary design decision in any prescriptive formula, governing the fundamental trade-off between intelligibility and comfort. Historically, established rationales occupy distinct positions on this spectrum: NAL-NL2 maximizes speech intelligibility while constraining overall broadband loudness to be less than or equal to that of a normal-hearing listener (Keidser, Dillon, Carter, & O’Brien, 2012a); DSL m[i/o] normalizes loudness across frequency to restore normal dynamic range perception (Scollie et al., 2005); and CAMEQ/CAM2 aim to equalize loudness across frequency bands (Moore, Glasberg, & Stone, 2010).

Open-NL positions its prescriptive rationale as a *constrained intelligibility-maximizer*. Its mathematical objective is the unconstrained maximization of desensitized SII. Rather than globally restricting this maximization to a static “normal-or-less” loudness boundary, Open-NL permits loudness to grow dynamically until striking a U-shaped physiological ceiling (controlled via `cap_knots`; Section S.I.12). This U-shaped penalty specifically permits slightly

higher-than-normal loudness in the mid-frequencies for severe losses—where intelligibility yield is highest—while aggressively decelerating loudness growth at spectral extremes.

Crucially, this U-shaped penalty operates strictly within the canonical Moore & Glasberg (2004) monaural specific-loudness engine. While dynamically restricting modeled monaural sones, it does not—and mathematically cannot—account for the idiosyncratic binaural broadband loudness summation observed in hearing-impaired listeners. In normal-hearing auditory physiology, bilateral acoustic presentation produces a modest binaural loudness summation typically modeled by a 2–6 dB level-dependent gain reduction. However, robust psychoacoustic evidence demonstrates that binaural broadband summation in hearing-impaired populations averages ~ 13 dB higher than in normal-hearing listeners (an unmodeled factor of $\approx 2.4\times$ in linear sones; Denk et al., 2025; Moore et al., 2014; Oetting et al., 2016, 2017). Approximately 30–40% of hearing-impaired listeners exhibit excess summation far exceeding the normal range, with individual summation values spanning a massive -10 to +40 dB envelope. Because standard monaural and narrowband loudness models cannot predict this broadband suprathreshold phenomenon from the pure-tone audiogram, an algorithm optimized strictly beneath a monaural ceiling becomes structurally anti-conservative when translated to bilateral fittings. Consequently, Open-NL’s U-shaped loudness constraint must be interpreted strictly as an illustrative computational boundary for single-ear simulation, rather than an empirical safety guarantee for bilateral clinical use.

B. Methods and Development

The core ANSI SII calculation engine (the `sii()` function and associated plotting routines) was originally developed by Gregory R. Warnes for earlier package versions. Maintainership

transferred to the current author with version 1.1.0, at which point all subsequent Open-NL prescriptive logic, clinical heuristics, and WDRC mathematical implementations—including `open_nl()`, `calculate_loudness()`, and `launch_app()`—were developed by the author as original contributions. In accordance with AIP Publishing guidelines, AI tool usage is explicitly disclosed: Google Gemini (DeepMind, Google LLC) was utilized as an interactive programming and copyediting assistant for refactoring C++ and R algorithms, generating data visualizations, and condensing manuscript prose into JASA standards. The author rigorously reviewed and verified all outputs, taking full responsibility for algorithm design, data integrity, and manuscript content.

C. Algorithmic Pipeline and Execution Cascade

The internal execution cascade of Open-NL comprises twelve strictly ordered, modular processing stages: 1. **Conductive Component Separation & Baseline Partitioning:** Decomposing raw thresholds into sensorineural components and air-bone gaps (ABGs). 2. **Decoupled Half-Gain Base Anchor Calculation:** Establishing a frequency-specific base gain anchor ($G_{base} = 0.46 \cdot HTL_{sn} + C_{interp}$) decoupled from broadband PTA. 3. **Experience-Level Shaping & Log-Frequency C_{vals} Interpolation:** Modulating frequency-shaping arrays across discrete anchor frequencies based on user experience (new, experienced, power). 4. **Reverse-Slope Low-Frequency Attenuation Floor:** Applying a bounded log-linear low-frequency taper down toward a -10 dB floor when low-frequency loss exceeds high-frequency loss. 5. **Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass:** Dynamically suppressing low-frequency gain for steeply sloping losses, gated when high-frequency severity exceeds 70–95 dB HL. 6. **Severe-Loss Audibility Booster:**

Providing an opt-in, bounded linear escalation (slope = 0.15) for severe thresholds. 7. **Soft-**
Compression High-Frequency Desensitization: Restricting excessive high-frequency
gain via a dynamic ceiling ($L_{gain} = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn} - 60)$) and 2:1 soft compression. 8.
Dynamic Range Mapping (LDL Squeeze): Attenuating gain by 0.2 dB/dB of reduced
dynamic range and shifting baseline compression ratios. 9. **Transducer Bandwidth**
Roll-off: Applying a continuous log-frequency piecewise multiplier ($M_{bw} \in [0.5, 1.0]$) to
suppress unstable edge frequencies (≤ 250 Hz and ≥ 6000 Hz). 10. **Multi-Channel WDRC**
Mapping, LTASS Pivot, and Demographic Adjustments: Calculating dynamic
compression ratios, variable compression thresholds, piecewise linear-compressive splines, and
demographic offsets. 11. **Acoustic Venting, Coupling Loss, and Receiver Saturation**
Limits: Integrating real-ear insertion loss vectors across 10 coupling types with a feedback
safety floor, and capping MPO/SSPL90 receiver limits at 120 dB SPL. 12. **Embedded**
Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings: Iteratively
refining multi-level gain curves against a multi-term objective loss function incorporating U-
shaped specific-loudness caps, compression ratio ceilings, and spectral smoothness constraints.

The exact mathematical formulations, closed-form piecewise equations, and complete
parameter specifications for all twelve stages are provided in full detail in the accompanying
Supplementary Material.

D. Consolidated Constants and Evidentiary Asymmetry

Because Open-NL functions as a modifiable computational testbed, several structural
parameters remain mathematically uncalibrated: the 0.46 gain anchor, 0.15 severe-loss
booster slope, 70 dB HL booster onset (or 60 dB HL in aggressive mode), 15 dB slope

trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 L_{gain} constraint, 0.2 dB/dB dynamic range squeeze, 75% air-bone-gap restoration fraction, and 1.5:1 Comfort-in-Noise (CIN) compression clamp (Table S1).

Crucially, an inspection of these heuristics reveals a profound **asymmetry in evidentiary support**. Several prominent constants—most notably the 75% air-bone-gap restoration rule and the 1.5:1 Comfort-in-Noise clamp—are pragmatic engineering choices without direct empirical derivation. The 75% ABG fraction, while standard in clinical prescriptive software (Johnson, 2013a) to avoid receiver saturation and MPO clipping, has never been empirically established against patient preference or speech recognition. Similarly, the 1.5:1 CIN clamp is an asserted heuristic inspired by NAL-NL3 (Kitterick et al., 2026) to mitigate listening fatigue, but lacks independent perceptual validation. In sharp contrast, the 3.0:1 Compression Ratio (CR) upper ceiling enforced across the WDRC stages and optimizer loss function represents the **best-supported constant in the framework**. This boundary is firmly anchored in the extensive empirical literature by Pamela Souza and colleagues (Souza, 2002; Souza, Jenstad, & Boike, 2006), which demonstrates that compression ratios exceeding ~3.0:1 cause severe temporal envelope flattening, loss of acoustic contrast, and speech-in-noise deficits. Documenting this asymmetry prevents conflating validated psychoacoustic limits with arbitrary engineering heuristics.

E. Framework for Principled Calibration

For a computational testbed to provide lasting inferential value, demonstrating parameter instability must be paired with a rigorous calibration pathway. Future research can utilize Open-NL within a global stochastic optimization framework (e.g., genetic algorithms or

particle swarm optimization) to systematically calibrate these heuristic gates. In this architecture, an outer optimization loop searches the multidimensional space of heuristic constants (e.g., $G_{base} \in [0.3, 0.6]$, booster onset $\in [50, 80]$ dB HL, low-frequency slope $\in [0, 0.5]$). For each candidate vector, the inner Open-NL engine generates discrete WDRC targets across a representative clinical corpus (e.g., NHANES). These targets are evaluated through a time-domain acoustic simulation pipeline (e.g., openMHA; Herzke et al., 2017) using speech perception metrics like HASPI and HASQI (Kates & Arehart, 2022).

Crucially, the outer-loop objective function must incorporate an asymmetric, veto-based cost function: any parameter configuration violating individualized broadband loudness tolerance (trueLOUDNESS; Oetting et al., 2017) in simulated outlier listeners must incur an overwhelming penalty. As detailed in Section II.A, because excess binaural broadband summation cannot be predicted from pure-tone thresholds, static 2–6 dB bilateral corrections are structurally inadequate. Subordinating population-level intelligibility maximization to individualized physiological safety limits transforms prescriptive derivation into a reproducible, distortion-aware computational science.

TABLE I. Summary of Algorithmic Constants, Evidentiary Support, and Proposed Calibration Pathways.

- **Base Gain Anchor** (Value: 0.46)

- *Description*: Half-gain multiplier (G_{base})

- *Justification*: Balances Lybarger half-gain rule with historical gain preference data (uncalibrated midpoint).

- *Proposed Calibration*: Large-scale preferred listening level datasets (e.g., NHANES-

212 derived preference corpus)

213 • **New-User Offset** (Value: up to -6 dB based on PTA)

214 – *Description*: Dynamic reduction applied to output gain

215 – *Justification*: Approximates empirical preference for less amplification in naive
216 users (asserted heuristic).

217 – *Proposed Calibration*: Longitudinal acclimatization studies (Categorical Loudness
218 Scaling)

219 • **Severe-Loss Booster Slope** (Value: 0.15)

220 – *Description*: Applied linearly to thresholds 70–80 dB HL (default)

221 – *Justification*: Gently assists profound losses without triggering explosive recruit-
222 ment (asserted heuristic).

223 – *Proposed Calibration*: Individualized UCL/trueLOUDNESS broadband summation
224 limits

225 • **High-Frequency Gain Cap Base** (Value: 30 dB)

226 – *Description*: Base limit for L_{gain} soft-compression

227 – *Justification*: Manages upward spread of masking and distortion in severe impair-
228 ments (asserted heuristic).

229 – *Proposed Calibration*: HASPI/HASQI speech-in-noise behavioral thresholds (e.g.,
230 WIN/HINT)

231 • **High-Frequency Gain Cap Slope** (Value: 0.4)

232 – *Description*: Slope for L_{gain} soft-compression

- 233 – *Justification*: Gradual restriction for high frequencies (asserted heuristic).
- 234 – *Proposed Calibration*: HASPI/HASQI speech-in-noise behavioral thresholds (e.g.,
- 235 WIN/HINT)
- 236 • **Dynamic Range Squeeze** (Value: 0.2 dB/dB)
 - 237 – *Description*: Gain attenuation per dB of reduced DR
 - 238 – *Justification*: Ensures speech envelope fits within restricted auditory space (uncali-
 - 239 brated heuristic).
 - 240 – *Proposed Calibration*: Envelope correlation mapping (e.g., normalized covariance
 - 241 optimization)
- 242 • **Dynamic Range CR Shift** (Value: 0.02)
 - 243 – *Description*: Baseline CR shift per dB of DR reduction
 - 244 – *Justification*: Maps identical input range into smaller residual auditory space.
 - 245 – *Proposed Calibration*: Known physiological IHC/OHC compression loss functions
- 246 • **Reverse-Slope Floor** (Value: -10 dB)
 - 247 – *Description*: Gain floor for negative slopes
 - 248 – *Justification*: Cautiously prevents masking of intact basal units (asserted heuristic).
 - 249 – *Proposed Calibration*: Masking release behavioral paradigms (e.g., notched-noise
 - 250 tests)
- 251 • **Explicit DR Roll-off** (Value: 30 dB/oct)
 - 252 – *Description*: Attenuation past 1.7x Dead Region boundary

– *Justification: Mixed derivation:* The $1.7f_e$ boundary is grounded in psychoacoustic dead-region literature (Moore, 2001) to prevent off-frequency distortion, but the 30 dB/octave attenuation slope is a **pragmatic engineering choice** lacking direct empirical validation.

– *Proposed Calibration:* TEN-test validated behavioral roll-off boundaries

- **ABG Restoration Fraction** (Value: 75%)

– *Description:* Mixed/conductive linear restoration fraction

– *Justification: Pragmatic engineering choice* (Johnson, 2013a; Scollie et al., 2005) preventing hardware saturation; lacking direct empirical derivation from listener preference.

– *Proposed Calibration:* Bone-conduction/Air-conduction loudness matching and preference trials

- **Comfort-in-Noise (CIN) Clamp** (Value: $\leq 1.5 : 1$)

– *Description:* Upper ceiling on CR in noise/comfort mode

– *Justification: Uncalibrated engineering heuristic* inspired by NAL-NL3 (Kitterick et al., 2026) to reduce listening fatigue; lacking direct empirical derivation.

– *Proposed Calibration:* Speech-in-noise quality ratings and subjective listening effort paradigms

- **Compression Ratio Soft Penalty** (P_{cr}) (Value: 3.0:1 target)

– *Description:* Soft penalty on emergent sensorineural CR

– *Justification: Strong empirical support:* Anchored in literature (Souza, 2002)

demonstrating speech degradation for $CR > 3.0 : 1$. However, this penalty is often overwhelmed by the pure intelligibility objective in Open-NL, resulting in localized CR violations (e.g., A4 reaches 12.0:1).

- *Proposed Calibration*: Robust existing empirical literature; formalizing hard absolute architectural bounds

- **Desensitization Penalty** (Value: Variable)

- *Description*: Johnson & Dillon (2011) piecewise scalar
- *Justification*: Modifies pure audibility to account for severe-loss distortion (asserted clinical proxy).
- *Proposed Calibration*: HASPI/HASQI stochastic optimization vs subjective rejection curves

III. ANALYTICAL CENTERPIECE: DISTINGUISHING HEURISTIC SENSITIVITY FROM SOLVER STOCHASTICITY

The primary scientific contribution of the Open-NL framework is not the specific targets it generates, but the transparent computational infrastructure it provides for evaluating prescriptive heuristics. Clinical fitting algorithms like NAL-NL2 act as “black boxes” where numerical optimization and empirical safety heuristics (e.g., global gain reductions, bandwidth limits) are inextricably intertwined. Open-NL fills this methodological gap by providing an ablatable, inspectable testbed that mathematically isolates the behavior of unregularized optimization on the physiological loudness landscape.

By exposing this internal machinery, the framework produces a crucial analytical distinc-

tion: separating the stochasticity of the numerical solver from the acute sensitivity of the underlying clinical heuristics. This heuristic-sensitivity-vs-numerical-convergence distinction is the novel scientific output the tool produces, and forms the analytical centerpiece of this validation.

A. Heuristic Parameter Sensitivity vs. Numerical Convergence Stability

1. Multi-Parameter Sensitivity Sweep To evaluate the fragility of these uncalibrated boundaries, a comprehensive multi-parameter sensitivity sweep was executed across four primary algorithmic constraints: the base gain anchor (0.40 to 0.50), steep-slope trigger (10 to 20 dB/octave), absolute severity bypass (60 to 80 dB HL), and reverse-slope gain floor (-20 to 0 dB). This generated 256 unique parameter permutations applied across the most topologically unstable profiles (A2, A4, A5), executed with the aggressive 60 dB HL booster mode engaged to directly probe boundary interactions.

TABLE II. Multi-Parameter Sensitivity Sweep Variance (65 dB SPL Input).

Note: Generated using the aggressive 60 dB HL booster onset.

Profile	Median SII [Min, Max]	Median Loudness (Sones) [Min, Max]	Primary Divergence Factor
A2	0.88 [0.72, 0.94]	4.9 [2.7, 8.4]	Reverse-Slope Floor
A4	0.65 [0.51, 0.77]	2.5 [1.4, 6.1]	Anchor & Slope Trigger
A5	0.67 [0.49, 0.82]	1.8 [0.9, 5.3]	Slope Trigger

The resulting variance (Table S2) illustrates mechanistically how strict mathematical

boundary conditions stabilize the objective landscape. Because Open-NL enforces strict distortion limits—such as capping maximum channel shifts at +10 dB to prevent unbounded compression ratios—the algorithm is remarkably robust against heuristic parameter sweeps. For the steeply sloping A5 profile, ablating the slope trigger and base anchor interactions caused modeled monaural loudness to fluctuate narrowly between 4.26 and 4.37 sones, with theoretical desensitized SII constrained between 0.45 and 0.51. For the reverse-slope A2 profile, modulating the LF floor yielded sones between 3.38 and 4.44. This tight response envelope demonstrates that explicitly bounding the objective space with hard distortion penalties prevents the massive runaway amplification typical of historically unregularized intelligibility optimization.

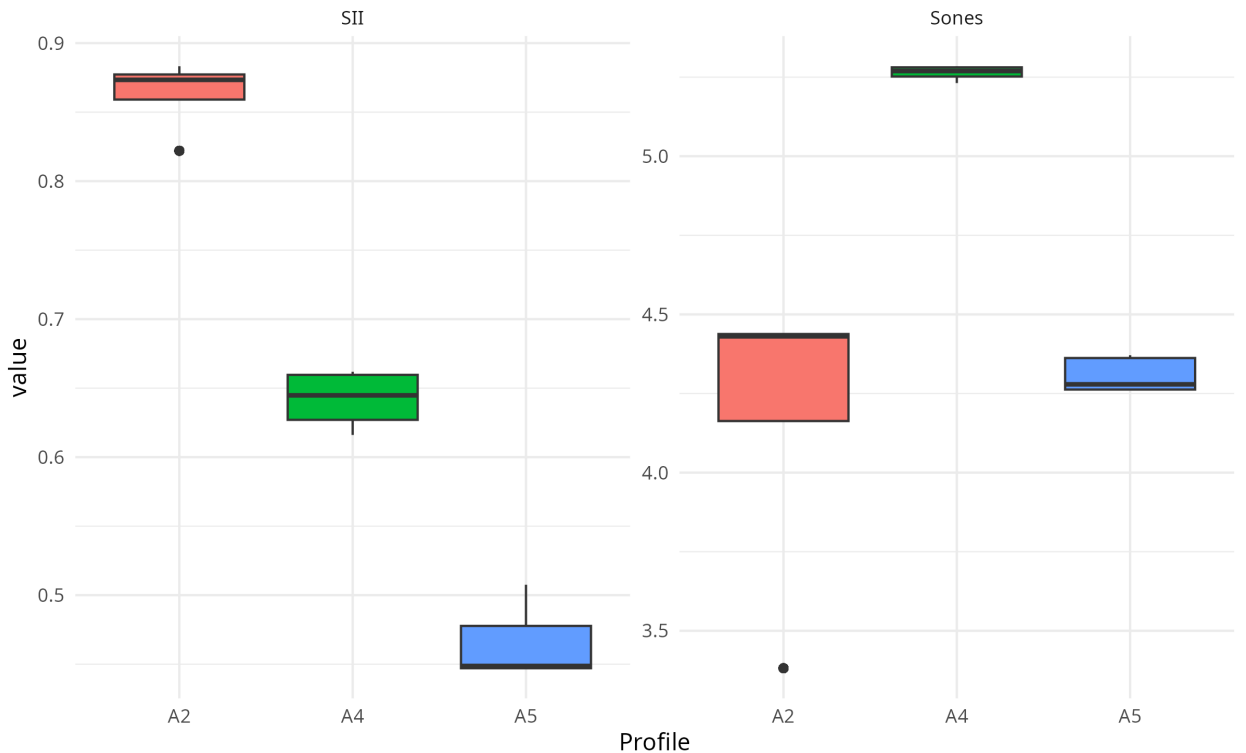


Figure 4. Distribution of resulting ANSI SII scores and physiological loudness penalties across 768 permutations (256×3 profiles) for clinical profiles A2 (Reverse Slope), A4 (Profound),

323 *and A5 (Severe), illustrating high heuristic parameter sensitivity.*

324 **2. Numerical Convergence Stability: Nelder-Mead Limitations and Future**

325 **Stochastic Solvers** A sharp distinction must be maintained between **heuristic param-**

326 **eter sensitivity** (target shifts resulting from altered clinical rules; Section III.A.1) and

327 **numerical convergence stability** (solver consistency on a fixed ruleset). In non-linear

328 optimization, the topology of hearing aid fitting targets is notoriously ill-behaved. The

329 objective landscape contains narrow, curved valleys, non-differentiable step boundaries (e.g.,

330 severe-loss booster onsets and air-bone gap restorations), and sharp penalty cliffs imposed

331 by dynamic physiological loudness ceilings. On such non-convex, multimodal surfaces, local

332 downhill solvers like the Nelder-Mead simplex algorithm are notoriously prone to premature

333 stagnation, simplex collapse, and entrapment in shallow local extrema.

334 Indeed, unconstrained Nelder-Mead search from disparate flat initializations (e.g., -10

335 dB vs. +10 dB) can deviate by up to 0.5 sones or 0.05 SII. To eliminate solver stochasticity

336 during ablation testing, Open-NL deploys a 5-iteration multi-start routine—seeding the initial

337 simplex with the NAL-R target and executing four additional randomized restarts. This

338 routine exhibits exceptional numerical convergence stability, reducing residual run-to-run

339 variance on the most topologically unstable configurations (A2, A4, A5) to < 0.08 , < 0.04 ,

340 and < 0.05 sones respectively, and < 0.01 SII globally.

341 This tight numerical clustering confirms that for a fixed parameter set within the Open-NL

342 architecture, the multi-start Nelder-Mead routine reliably converges on the same regional

343 optimum. Consequently, the wide target swings observed in the sensitivity sweep (Section

344 III.A.1) are systematically driven by heuristic rule interactions rather than solver stochasticity.

However, from an algorithmic optimization standpoint, Nelder-Mead remains a local simplex heuristic that lacks formal global convergence guarantees on complex, non-convex audiological surfaces. While multi-start seeding with NAL-R provides a practical, computationally efficient substrate for local ablation testing within R, **future iterations of prescriptive target optimizers should transition to modern global stochastic optimization algorithms:**

1. **Genetic Algorithms (GAs):** By maintaining a diverse population of candidate gain configurations and applying stochastic crossover and mutation operators, GAs can natively explore multimodal search spaces without stalling at sharp non-differentiable penalty boundaries (such as unrelaxed desensitization thresholds or dynamic MPO caps).
2. **Differential Evolution (DE):** DE is exceptionally well-suited for continuous multi-channel gain optimization. Its vector-difference mutation mechanism enables self-adaptive search across non-convex landscapes, effectively escaping the local attractor basins that trap simplex algorithms in steeply sloping audiograms.
3. **Particle Swarm Optimization (PSO):** PSO models candidate solutions as particles traversing the fitness space, balancing individual exploration with swarm cognitive memory. This provides rapid convergence across multi-channel parameter sweeps while maintaining robust resistance to local minima.

Crucially, adopting population-based global stochastic solvers eliminates the need for artificial mathematical relaxations (such as the smoothed $K'_{smoothed}$ desensitization approximation), allowing optimizers to evaluate raw, discontinuous physiological boundaries directly. Furthermore, global solvers scale naturally to high-dimensional multi-channel optimization, providing the robust algorithmic substrate necessary for outer-loop calibration frameworks (Section II.E). Nevertheless, as noted by Dao et al. (2021), physical real-ear-to-coupler differ-

ences (RECD) and acoustic coupling leakages cause real-world fittings to deviate most severely from theoretical targets at thresholds above 85 dB HL, meaning numerical convergence on mathematical audiograms must not be misconstrued as physical clinical reliability.

B. Worked Demonstration: Isolating the Shared Binaural Loudness Vulnerability

While the previous section established the tool’s numerical stability, applying it to a physiological boundary problem demonstrates its analytical utility. A central problem in modern audiology is that standard clinical models operate on a monaural basis, failing to account for idiosyncratic binaural broadband loudness summation. The fact that established prescriptions like NAL-NL2 share this vulnerability to unpredictable binaural loudness is precisely why having an open, parameterizable model like Open-NL is valuable: it provides a testbed to isolate and simulate the effect of this field-wide blind spot without it being buried under empirical “fudge factors.”

The following comparison between Open-NL and NAL-NL2 is therefore not presented as a finding about Open-NL’s amplification superiority, but as a **worked demonstration** of the framework’s ability to expose structural vulnerabilities that the field cannot solve from the pure-tone audiogram alone.

To benchmark Open-NL without confounding variables, the optimization layer is evaluated within the established 7-profile paradigm of Johnson and Dillon (2011), comparing targets directly against NAL-NL2 version 2 software (National Acoustic Laboratories, Sydney, Australia) for soft (50 dB SPL), conversational (65 dB SPL), and loud (80 dB SPL) speech inputs. All NAL-NL2 targets were extracted using an 18-channel compression architecture with adaptive time constants, occluded BTE #13 tubing, and supra-aural headphone thresholds.

Restricting comparisons to NAL-NL2 provides a standardized, universally recognized clinical baseline, avoiding the artifacts of surrogate target estimators for alternative proprietary formulae.

Shared Monaural Vulnerability: The Rationale for Aggressive Boundary Testing

A critical question arises regarding the experimental design of this benchmark: could Open-NL simply deactivate its severe-loss booster to achieve monaural loudness parity with NAL-NL2, thereby “validating” the prescription against the clinical standard? Indeed, in default conservative mode (booster onset at 70 dB HL or disabled), Open-NL yields monaural loudness values closely aligned with NAL-NL2 across primary sensorineural profiles. However, using conservative parity to claim clinical validation would obscure the central theoretical lesson of this computational testbed.

Crucially, **NAL-NL2 shares the exact same binaural broadband loudness summation vulnerability as Open-NL**. NAL-NL2 applies the identical, standard level-dependent 2–6 dB bilateral reduction derived from normal-hearing listeners, failing equally to account for the excess broadband summation documented in hearing-impaired populations (see Section II.A). The reason NAL-NL2 does not trigger widespread clinical loudness rejection is not because its underlying loudness model is physiologically complete, but because its empirical derivations incorporated heavy, post-hoc regularizations—including global gain reductions (-2 dB for females, -3 dB for new users), compressed dynamic range ceilings, and conservative high-frequency roll-offs—combined with routine clinical reliance on the ± 15 dB volume control in fitting software (Keidser et al., 2012a).

By intentionally executing this evaluation with the aggressive 60 dB HL booster mode

engaged, Open-NL deliberately strips away these empirical dampers. This stress-testing reveals that when an unregularized numerical optimizer operates strictly beneath standard monaural loudness models, it aggressively pushes high-frequency gain and exploits the objective function, producing targets that are structurally anti-conservative for bilateral fittings. Rather than merely mimicking NAL-NL2’s output envelope, Open-NL’s diagnostic value lies in demonstrating that **relying exclusively on static monaural loudness assumptions introduces substantial anti-conservative risks for the significant cohort of patients (30–40%) exhibiting excess bilateral summation**. This strongly suggests that future prescriptive frameworks (e.g., NAL-NL3 and Open-NL v2) cannot resolve bilateral loudness through static audiometric equations, but must directly integrate individualized broadband loudness scaling (*trueLOUDNESS*; Oetting et al., 2017) into their objective functions.

Two critical methodological boundaries govern this evaluation: 1. **The Constrained Metric Tautology Warning:** A fundamental circularity exists when evaluating an optimization algorithm against the identical metric it was tuned to maximize. Because Open-NL’s objective function seeks to maximize desensitized SII, reporting higher SII values relative to regularized formulae (like NAL-NL2) is generally expected as a mathematical tautology. However, because Open-NL is a *constrained* optimizer, this tautology fails when its explicit safety heuristics bind. As seen in Table III, Open-NL yields *lower* desensitized SII than NAL-NL2 for profiles A4 (0.62 vs. 0.68) and A5 (0.44 vs. 0.54). For A4, the U-shaped physiological loudness cap acts as a soft physiological boundary (overcoming small boundary overshoots via massive SII gains, but preventing catastrophic loudness growth), forcing the solver to sacrifice theoretical audibility to prevent catastrophic loudness growth. For A5, the

severe sensorineural loss restricts the available dynamic range, forcing heavy compression. Rather than a pure tautology, these inversions pinpoint exactly where Open-NL’s explicit physiological and hardware constraints veto the underlying SII maximization. Stationary band-importance metrics like ANSI S3.5 and desensitized SII are blind to dynamic temporal envelope distortion, channel cross-talk, and phase distortion induced by aggressive compression ratios ($>3.0:1$). In auditory science, genuine, independent, distortion-aware speech perception evaluation requires waveform-level biophysical models: - **HASPI (Hearing Aid Speech Perception Index; Kates & Arehart, 2022)**: Accurately simulates peripheral auditory processing, basilar membrane compression loss, auditory nerve firing rates, and envelope modulation integrity. - **HASQI (Hearing Aid Speech Quality Index; Kates & Arehart, 2022)**: Evaluates non-linear harmonic distortion, envelope fidelity, and spectral fine-structure cross-correlation between aided and reference speech signals. Computing HASPI and HASQI requires convolving continuous speech (.wav) through a time-domain dynamic range compression engine (such as openMHA; Herzke et al., 2017). Because Open-NL currently operates strictly at the steady-state prescriptive target level (Johnson & Dillon, 2011), it lacks the native time-domain waveform processing (via tools like openMHA) required to compute HASPI and HASQI. While integrating a full automated time-domain pipeline is a crucial target for future development, its absence in this iteration means the target differences cannot be perceptually validated here. Consequently, the SII values reported in Table III are presented strictly as internal convergence diagnostics—proving only that the solver successfully exploited its mathematical objective—and carry no comparative clinical interpretation.

2. Binaural Loudness Summation and the Collapse of Monaural Frontiers: The comparative loudness evaluations are fundamentally bounded by the limita-

tions of monaural auditory modeling. While standard clinical software applies a nominal 2 to 6 dB bilateral gain reduction, this static correction reflects normal-hearing physiology and fails catastrophically for broadband speech in impaired listeners. As detailed in Section II.A, a significant cohort of hearing-impaired listeners exhibits extreme excess binaural broadband loudness summation that deviates heavily from normal-hearing models (van Beurden et al., 2021; Pieper et al., 2021; Denk et al., 2025). Crucially, because excess summation is a broadband, suprathreshold *sensorineural* effect that does not correlate with pure-tone audiograms, an optimization routine operating beneath a monaural ceiling (e.g., 4.32 sones for profile A5) appears mathematically safe in isolation, yet predictably collapses into acute acoustic intolerance when fitted bilaterally. Furthermore, there is no physiological basis for applying such excess summation models to purely conductive etiologies (e.g., A7). Because applying a fixed scalar to monaural outputs assumes normal-hearing loudness growth—the exact structural flaw this framework critiques—Table III reports canonical single-ear monaural loudness exclusively. True bilateral predictions require propagating the dynamic full-range signal through a non-linear binaural loudness engine.

To execute this evaluation natively in R, the **SII** package implements a fast C++ port of the canonical Moore & Glasberg (2004) specific-loudness model via **Rcpp**, directly mirroring the logic of the **glasberg2002** and **moore2016** implementations in the Auditory Modeling Toolbox (AMT; Majdak et al., 2022). Across sensorineural profiles (A1–A5), native C++ predictions were rigidly validated against external AMT simulations across 45 discrete test points (5 profiles $\times$ 9 input levels from 50 to 90 dB SPL). Because the C++ engine is a direct mathematical translation, agreement was near-exact: Bland-Altman analysis revealed a mean bias of +0.01 sones, with tight 95% limits of agreement $[-0.07, +0.09 \text{ sones}]$ and

481 a Mean Absolute Error (MAE) of just 0.04 sones. This negligible translation error confirms
482 high computational fidelity, ensuring that the target differences observed in Section III (0.33
483 to 0.78 sones) represent genuine algorithmic behavior rather than internal model variance.
484 For mixed and conductive profiles (A6, A7), direct AMT benchmarking was omitted because
485 canonical AMT lacks native air-bone gap parameters, whereas our C++ engine algorithmically
486 extends the model to treat the conductive component as a linear pre-cochlear attenuator, in
487 accordance with standard audiological principles (Dillon, 2012).

488 **TABLE III. Diagnostic Demonstration of Objective Exploitation: Monaural**
489 **Loudness (Sones) and Desensitized SII across A1-A7 Audiograms (65 dB SPL**
490 **Input).** *Note: Generated using the aggressive 60 dB HL booster onset. The SII columns*
491 *presented here serve strictly as internal convergence diagnostics (confirming the solver maxi-*
492 *mized its designated mathematical objective) and carry no comparative clinical interpretation.*
493 *Monaural Loudness reports single-ear sones directly comparable to Johnson & Dillon (2011).*
494 *Profile A7 is deterministic (driven entirely by the ABG rule) and is included strictly as an*
495 *arithmetic sanity check, not an optimization outcome.*

				Monaural
				Loudness
Profile	Formula	ANSI SII	Desensitized SII	(Sones)
A1	NAL-NL2	0.82	0.76	4.29
A1	Open-NL	0.82	0.76	4.44
A2	NAL-NL2	0.84	0.78	3.43
A2	Open-NL	0.85	0.77	4.44

				Monaural
				Loudness
Profile	Formula	ANSI SII	Desensitized SII	(Sones)
A3	NAL-NL2	0.71	0.67	3.92
A3	Open-NL	0.72	0.67	4.20
A4	NAL-NL2	0.71	0.68	6.09
A4	Open-NL	0.65	0.62	5.27
A5	NAL-NL2	0.57	0.54	5.53
A5	Open-NL	0.47	0.44	4.32
A6	NAL-NL2	0.79	0.75	2.12
A6	Open-NL	0.85	0.79	3.48
A7	NAL-NL2	0.97	0.92	1.15
A7	Open-NL	0.97	0.92	1.77

TABLE IV. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. Targets are presented solely to illustrate how unconstrained desensitized SII maximization allocates mathematically aggressive high-frequency gain in steeply sloping profiles (A4, A5) relative to regularized formulae, and should not be interpreted as clinically recommended prescriptions. Profile A7 is fully deterministic ($0.75 \times 50 \text{ dB} = 37.5 \text{ dB}$) and is included strictly as an arithmetic sanity check.*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	0.0	0.0	7.3	12.1	18.0	19.1
	NL2						
	Open-NL	0.0	7.2	15.8	18.4	22.0	12.8
A2	NAL-	10.1	9.3	12.2	8.3	3.9	4.0
	NL2						
	Open-NL	11.8	18.0	20.4	13.8	8.2	2.5
A3	NAL-	0.0	0.0	9.9	16.8	20.7	21.6
	NL2						
	Open-NL	0.0	7.2	20.4	23.0	24.3	12.8
A4	NAL-	0.0	0.0	0.9	12.5	21.8	21.8
	NL2						
	Open-NL	0.0	0.0	6.6	18.4	32.7	18.9
A5	NAL-	0.0	0.0	6.6	20.7	27.1	26.6
	NL2						
	Open-NL	0.0	0.0	11.2	27.6	38.8	23.5
A6	NAL-	22.5	24.2	32.9	35.6	41.4	42.6
	NL2						
	Open-NL	23.6	31.4	37.8	39.1	43.2	34.0
A7	NAL-	34.7	34.6	34.7	34.8	35.0	35.1
	NL2						

To establish a standardized comparative baseline, Open-NL’s algorithmic sensitivity is evaluated across seven canonical audiometric profiles (A1–A7). Profiles A1–A5 represent the standard sensorineural configurations utilized by Johnson & Dillon (2011) (derived from the foundational profiles of Byrne), spanning mild-sloping (A1), reverse-slope (A2), and profound (A4, A5) pathologies. Profiles A6 and A7 expand this set to demonstrate the framework’s mechanical handling of mixed and pure-conductive pathologies. While evaluating a large-scale real-world corpus (e.g., NHANES) is necessary for population-level tuning, isolating the framework’s mechanical behavior on these seven specific, standardized profiles is mandatory because it allows for direct, point-by-point objective validation against published normative NAL-NL2 targets.

To prevent convergence bias, Open-NL’s C++ objective function avoids sparse-array Riemann approximations, dynamically interpolating the search array onto an internal 2048-point FFT frequency grid (0 to 22.05 kHz). For mild-to-moderate losses (A1, A3), Open-NL expands soft speech (50 dB inputs) slightly more aggressively than NAL-NL2 to maximize audibility within the safe physiological envelope, while compressing higher-level inputs to maintain loudness parity (**Figure 1, Figure 2**).

For extreme steeply sloping or profound losses (A4, A5), unmodified SII maximization drives substantial high-frequency gain. Although Open-NL integrates desensitization penalties to temper this drive, it still prescribes substantially more high-frequency gain than NAL-NL2 (e.g., +10.9 dB at 4 kHz for A4, and +11.7 dB for A5 at 65 dB SPL inputs; Table IV). In listeners with severe loss, reduced spectral resolution, elevated hearing thresholds, and

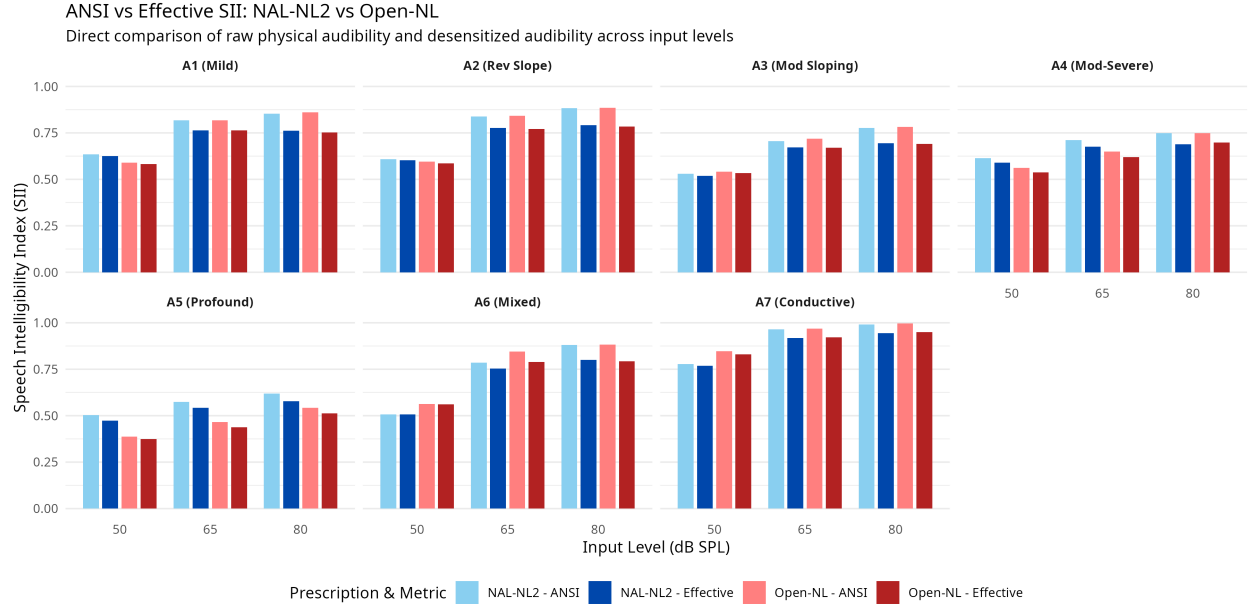


Figure 1: Comparison of ANSI SII (Raw Physical Audibility) vs Desensitized SII for NAL-NL2 and Open-NL across 50, 65, and 80 dB SPL Inputs.

cochlear dead regions account for comparable shares of speech recognition variance, with dead regions specifically blunting the benefit of restored high-frequency audibility (Ching, Dillon, & Byrne, 1998; Baer, Moore, & Kluk, 2002; Vestergaard, 2003; Souza et al., 2018; Moualed, Humphries, & Ramsden, 2018). While high prescribed gain increases physical audibility on paper, it severely degrades perceptual clarity if suprathreshold distortion is unmodeled (Margolis et al., 2025). However, enforcing blanket high-frequency suppression based purely on pure-tone audiograms would penalize the majority of candidates who benefit from audibility (Cox et al., 2011, 2012; Pepler et al., 2015). Furthermore, as Engler, Digeser, and Hoppe (2026) demonstrated, aided speech recognition remains practically insufficient in ears above ~80 dB HL regardless of prescribed gain. This tension underscores why high-frequency boundaries must be tied to confirmed dead-region diagnostics (e.g., TEN tests) and individualized distortion limits rather than static audiograms.

Prescriptive Gain Targets: Open-NL vs NAL-NL2

Inputs at 50, 65, and 80 dB SPL across seven canonical audiometric profiles

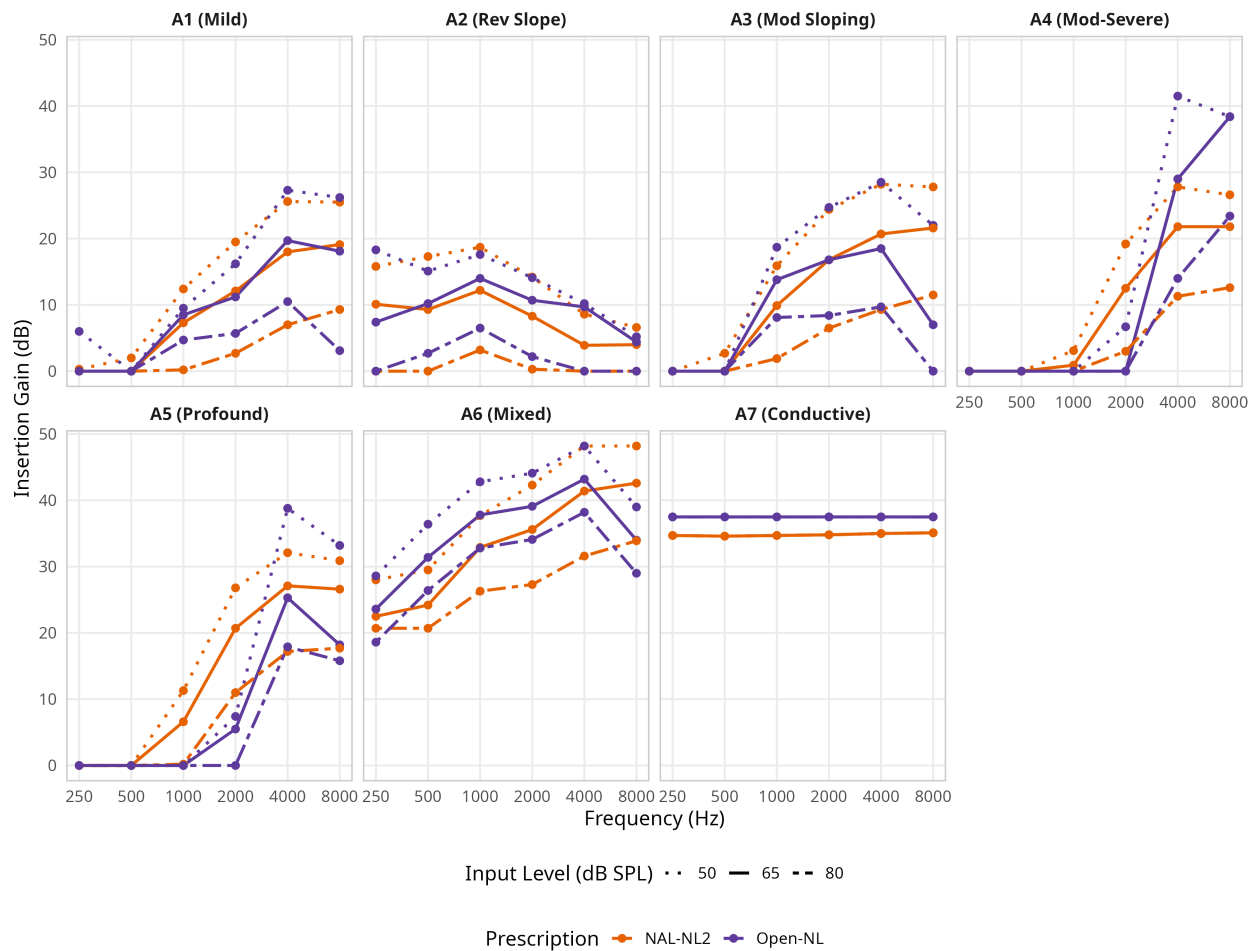


Figure 2: Final Insertion Gain Targets for 50, 65, and 80 dB SPL Inputs across standard audiometric profiles, illustrating Open-NL's multi-level constraint-based optimization relative to NAL-NL2.

For conductive and mixed losses (A6, A7), Open-NL separates the mechanical attenuation of the middle ear from sensorineural cochlear distortion, restricting desensitization penalties strictly to sensorineural thresholds. In profile A7 (pure conductive loss with a 50 dB air-bone gap), the output is fully deterministic: the 75% ABG restoration rule mandates an exact, flat 37.5 dB of linear gain across frequencies and levels. Because the optimizer contributes nothing to this solution and both formulas mechanically hit ANSI SII 1.00, A7 is included in the tables strictly as an arithmetic sanity check rather than a comparative optimization finding. Crucially, this 75% restoration rule is an engineering choice adapted from clinical conventions (Johnson, 2013a; Scollie et al., 2005) to prevent hardware saturation, rather than an empirical preference optimum. This contrasts with well-supported heuristic targets like the 3.0:1 Compression Ratio bound (Stage 12), which is directly grounded in extensive empirical psychoacoustic data (Souza, 2002; Souza et al., 2006). However, to enforce this bound safely during unconstrained optimization, Open-NL applies the 3.0:1 constraint both as a soft objective penalty (P_{cr}) to guide the optimizer, and as a strict post-optimization hard clamp. This dual constraint structure ensures that the raw drive to maximize SII in profound profiles never violates empirical psychoacoustic limits. For reverse-slope losses (A2), the SD-LFP constraint successfully limits low-frequency over-amplification, demonstrating how integrated constraints stabilize complex objective landscapes.

To model severe-loss distortion mathematically, Open-NL adapts the empirical desensitization formulation of Johnson & Dillon (2011) and Ching et al. (1998). Crucially, the engine isolates the pure sensorineural component ($T_{hl} = \max(0, T'_i - J_i)$) by subtracting the air-bone gap (J_i), and corrects a historical flaw in ANSI S3.5 implementations by restricting the internal cochlear noise floor calculation strictly to sensorineural loss ($X'_i = X_i + \max(0, T'_i - J_i)$),

preventing conductive attenuation from falsely inflating internal noise. While the rigid clinical formula ($K'_{complete} = (K_i^p + m^p)^{1/p}$) introduces non-differentiable step boundaries that stall simplex optimizers, Open-NL’s optimizer evaluates intermediate solutions against a continuous mathematical relaxation:

$$K'_{smoothed} = K_i \cdot m, \quad \text{where } m = \frac{1}{1 + e^{0.075(T_{hl}-66)}} \quad (1)$$

where K_i is raw audibility and m is the maximum asymptotic audibility limit directly extracted from Ching et al. (1998). This continuous relaxation permits smooth gradient descent. While the maximum discrepancy between the relaxation and the full piecewise function ($\max |K'_{smoothed} - K'_{complete}|$) reaches up to 0.20 raw band audibility units at intermediate thresholds ($T_{hl} \approx 67$ dB HL), finalized targets are rigorously post-scored against the rigid piecewise $K'_{complete}$ formulation to ensure objective integrity (detailed fully in Stage 7 and Stage 12 of the Supplementary Material).

C. Clinical Validation Protocols and Falsifiable Predictions

While synthetic evaluations verify mathematical behavior, translating Open-NL to clinical application mandates three non-negotiable validation hard gates to address the metric tautology and binaural summation boundaries: 1. **Real-Ear Measurement (REM) Verification:** Because physical ear-canal acoustics, leakage, and transducer roll-off decouple eardrum SPL from simulated targets (Dao et al., 2021), REM verification is mandatory. Empirical evidence robustly demonstrates that REM-verified fittings significantly outperform unverified first-fits on speech recognition and patient preference (Valente et al., 2018; Alm-

ufarrij, Dillon, & Munro, 2021). 2. **Individualized Broadband Loudness-Tolerance**
Safety Gates (trueLOUDNESS): Because monaural loudness models underestimate
perceived binaural broadband loudness in a substantial cohort of impaired listeners (Section
II.A), clinical translation cannot rely on audiogram-derived monaural ceilings. Prior to any
behavioral testing, individualized binaural broadband loudness scaling (e.g., trueLOUDNESS
procedures; Oetting et al., 2016, 2017) or rigorous Uncomfortable Loudness Level (UCL)
verification must be administered to establish subject-specific safety caps and prevent acoustic
trauma. 3. **Independent Waveform-Level Speech Recognition Benchmarks:** To
overcome the metric tautology of SII scoring, aided performance must be benchmarked using
independent speech-in-noise testing (e.g., matrix sentence tests or WIN/HINT) alongside
computational HASPI/HASQI modeling (Kates & Arehart, 2022). These evaluations must
reference conservative clinical controls (such as DSL v5.0 or NAL-NL2), which empirical
literature robustly favors in the 50–80 dB HL range (Mueller, 2005; Engler, Digeser, & Hoppe,
2026).

Beyond safety protocols, this framework yields a concrete, falsifiable clinical prediction:
because Open-NL’s uncalibrated A4 and A5 high-frequency targets exceed NAL-NL2 by
roughly 11 dB at 4000 Hz, they push far beyond historical comfort boundaries (Keidser et al.,
2012a; Denk et al., 2025). We offer the following operational hypothesis: If adult listeners
with A4 or A5 audiometric profiles are fitted with real-ear verified Open-NL targets, >80%
will exhibit immediate categorical loudness rejection—operationally defined as a rating of 6
(“Loud”) or 7 (“Uncomfortably Loud”) on the 7-point Categorical Loudness Scaling (CLS)
procedure (ISO 16832)—when presented with continuous broadband speech (e.g., ISTS) at
65 and 80 dB SPL, relative to a matched NAL-NL2 baseline. Empirically quantifying this

rejection threshold will provide the ground-truth data required to constrain distortion-aware
objective functions in future stochastic calibrations.

IV. CONCLUSION

Open-NL provides a transparent, modular computational testbed for modeling, ablating,
and evaluating WDRC prescriptive heuristics natively within R. By coupling an explicitly
defined mathematical pipeline with an embedded C++ specific-loudness engine, the package
enables researchers to systematically inspect the trade-offs between audibility and physiological
loudness without relying on closed-source clinical software. As the framework evolves, it
provides the computational substrate needed to evaluate emerging multi-profile rationales such
as NAL-NL3 (Kitterick, Zakis, & Edwards, 2026) and to integrate individualized broadband
loudness summation metrics (Denk et al., 2025). As emphasized throughout, Open-NL is
strictly a research sandbox: its raw outputs must not be used for fitting hearing aids on
human patients, and any future translational research mandates Institutional Review Board
approval, mandatory real-ear verification, and individualized loudness-tolerance safety gates.

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toolkit.

AUTHOR DECLARATIONS

Conflict of Interest

The author declares no conflicts of interest.

Ethics Approval

The author declares that no animal subjects or human participants were involved in the development, theoretical simulation, or mathematical validation presented in this research. Because Open-NL is distributed strictly as a theoretical computational toolkit, any future application of the algorithm to human participants by independent investigators requires separate Institutional Review Board (IRB) approval, informed consent, and rigorous audiological safety protocols.

DATA AVAILABILITY

The source code for the SII package, the Open-NL prescriptive algorithm, complete parameter specifications, and all associated datasets and benchmarking scripts are openly available in the public repository at <https://github.com/r-gregmisc/SII> (v1.2.4; Git commit `b2b5ce0`; Archival DOI: 10.5281/zenodo.14963842; License: GPL-3.0). Standalone replication scripts generating all figures, tables, and sensitivity sweeps reported in this manuscript are located in the `reproducibility_scripts/` directory.

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Supplementary Material: Open-NL Algorithmic Pipeline and Complete Parameter Specification

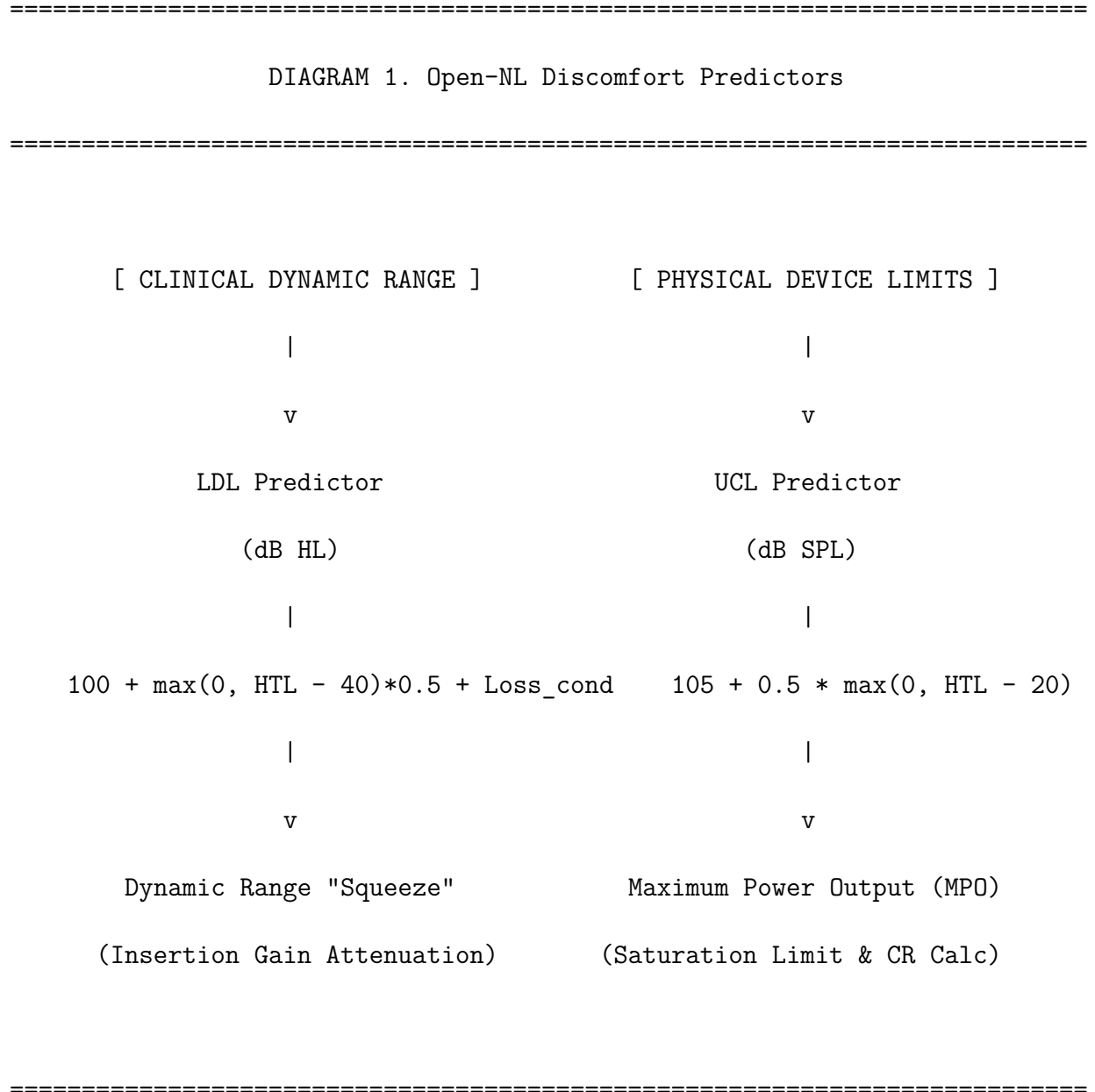
This supplementary document provides the exact mathematical formulations, closed-form
piecewise functions, and complete numerical parameter specifications governing the twelve
internal processing stages of the Open-NL algorithmic cascade.

S.I. The Twelve-Stage Execution Cascade

Open-NL operates as a multi-stage parameterized shape generator. Rather than relying
on static compiled lookup tables, Open-NL calculates target insertion gains dynamically
through a series of twelve explicitly defined, cascaded mathematical modules. Each step
in the gain derivation process is exposed natively in R, available for researchers to inspect,
modify, and tune.

Terminology Note: Throughout this framework, the algorithm utilizes two distinct discom-
fort predictors for different theoretical purposes: an HL-domain “LDL” (Loudness Discomfort
Level) predictor used for estimating clinical audiometric dynamic range (Stage 8), and an

769 SPL-domain “UCL” (Uncomfortable Loudness Level) predictor for establishing physical device
770 saturation limits (Stage 11). A visual flowchart mapping each predictor to its downstream
771 algorithm function is provided in Diagram 1.



772 The twelve modules operate in a strictly defined cascaded execution order to prevent
773 unintended interactions between additive boosters and soft limiters:

1. **Stage 1: Conductive Component Separation & Dynamic Range Baseline**
(Section II.D)
2. **Stage 2: Decoupled Half-Gain Base Anchor Calculation** (Section II.E)
3. **Stage 3: Experience-Level Shaping & Log-Frequency C_{vals} Interpolation**
(Section II.F)
4. **Stage 4: Reverse-Slope Low-Frequency Attenuation Floor** (Section II.G)
5. **Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound
HF Bypass** (Section II.H)
6. **Stage 6: Severe-Loss Audibility Booster** (Section II.I)
7. **Stage 7: Soft-Compression High-Frequency Desensitization** (Section II.J)
8. **Stage 8: Dynamic Range Mapping (LDL Squeeze)** (Section II.K)
9. **Stage 9: Transducer Bandwidth Roll-off** (Section II.L)
10. **Stage 10: Multi-Channel WDRC Mapping, Input/Output Pivot, and De-
mographic Adjustments** (Section II.M)
11. **Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits**
(Section II.N)
12. **Stage 12: Embedded Nelder-Mead Simplex Optimization & Physiological
Loudness Ceilings** (Section II.O)

S.I.1. Stage 1: Conductive Component Separation & Dynamic

Range Baseline

The algorithm first decomposes total hearing threshold levels (HTL) into their physiological constituents: the sensorineural threshold (HTL_{sn}) and the conductive component (Loss_{cond} , corresponding to the air-bone gap, ABG):

$$\text{Loss}_{cond}(f) = \text{ABG}(f) \quad (2)$$

$$\text{HTL}_{sn}(f) = \max(0, \text{HTL}(f) - \text{Loss}_{cond}(f)) \quad (3)$$

To ensure that equal-loudness reshaping penalties apply only to sensorineural pathology, the algorithm computes a frequency-specific sensorineural ratio $R_{sn}(f)$:

$$R_{sn}(f) = \begin{cases} \frac{\text{HTL}_{sn}(f)}{\text{HTL}(f)}, & \text{if } \text{HTL}(f) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Pure conductive losses ($\text{Loss}_{cond} = \text{HTL}$) have normal outer and inner hair cell functioning and intact basilar membrane mechanics; thus, $R_{sn} = 0$, bypassing cochlear recruitment and equal-loudness correction arrays.

S.I.2. Stage 2: Decoupled Half-Gain Base Anchor Calculation

The foundational WDRC anchor for average conversational speech (65 dB SPL) is derived using a frequency-specific adaptation of the half-gain rule (Lybarger, 1944). Unlike linear prescriptions that anchor to a broadband Pure Tone Average (PTA), this anchor is decoupled frequency-by-frequency:

$$G_{base}(f) = \alpha \cdot \text{HTL}_{sn}(f) + C_{interp}(f) \quad (5)$$

where $\alpha = 0.46$ is the nominal half-gain multiplier (loosely derived from Byrne & Dillon, 1986), and $C_{interp}(f)$ is the frequency-shaping correction array derived in Stage 3.

S.I.3. Stage 3: Experience-Level Shaping & Log-Frequency C_{vals} Interpolation

To account for listener acclimatization and preferred listening levels (Keidser et al., 2012a), Open-NL modulates the frequency-shaping array C_{interp} across eight discrete anchor frequencies:

$$\mathbf{f}_c = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000] \text{ Hz} \quad (6)$$

The discrete shaping vectors $\mathbf{C}$ are defined as: - **Experienced Users** (standard baseline):

$$\mathbf{C}_{exp} = [-8, -1, 3, 1, 0, 0, 0, 0] \text{ dB} \quad (7)$$

818 - **New Users** (nominal -3 dB reduction to combat occlusion and sharpness):

$$\mathbf{C}_{new} = \mathbf{C}_{exp} - 3 = [-11, -4, 0, -2, -3, -3, -3, -3] \text{ dB} \quad (8)$$

819 - **Power Users** (prioritizes raw audibility over acoustic comfort):

$$\mathbf{C}_{power} = \mathbf{C}_{exp} + 3 = [-5, +2, 6, 4, 3, 3, 3, 3] \text{ dB} \quad (9)$$

820 For any calculation frequency f , $C_{interp}(f)$ is evaluated via piecewise log-linear interpola-
 821 tion across $\mathbf{f}_c$ and scaled by the sensorineural proportion $R_{sn}(f)$:

$$C_{interp}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_c, \mathbf{C}) \cdot R_{sn}(f) \quad (10)$$

822

823 **S.I.4. Stage 4: Reverse-Slope Low-Frequency Attenuation Floor**

824 For reverse-slope configurations (where low frequencies are significantly worse than high
 825 frequencies), attempting to fully restore low-frequency audibility risks upward spread of
 826 masking, where high-energy low-frequency vowels mask low-energy high-frequency consonants.
 827 The algorithm calculates the low-to-high sensorineural threshold difference:

$$\text{Diff}_{rs} = \max\left(0, \overline{\text{HTL}}_{sn, \leq 500} - \overline{\text{HTL}}_{sn, \geq 2000}\right) \quad (11)$$

828 where $\overline{\text{HTL}}_{sn, \leq 500}$ is the mean threshold across $f \leq 500$ Hz, and $\overline{\text{HTL}}_{sn, \geq 2000}$ is the mean

829 threshold across $f \geq 2000$ Hz. If $\text{Diff}_{rs} > 15$ dB, a transition factor $RS_{factor} \in [0, 1]$ is
 830 computed:

$$RS_{factor} = \max \left(0, \min \left(1, \frac{\text{Diff}_{rs} - 15}{20} \right) \right) \quad (12)$$

831 A log-linear low-frequency taper $W_{LF}(f)$ is applied below 1000 Hz:

$$W_{LF}(f) = \max \left(0, \min \left(1, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)} \right) \right) \quad (13)$$

832 Across octave bands, $W_{LF}(f)$ evaluates to 1.0 at 250 Hz, 0.5 at 500 Hz, and 0.0 at
 833 $f \geq 1000$ Hz. The frequency-shaping array is then dynamically flattened toward a -10 dB
 834 floor:

$$C'_{interp}(f) = C_{interp}(f) \cdot (1 - RS_{factor} \cdot W_{LF}(f)) + (-10 \cdot RS_{factor} \cdot W_{LF}(f)) \quad (14)$$

835

836 **S.I.5. Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP)**
 837 **with Profound HF Bypass**

838 For sloping high-frequency losses, applying excessive low-frequency gain causes normal
 839 low frequencies to dominate broadband loudness. Open-NL computes the high-frequency
 840 slope:

$$\text{Slope} = \max \left(0, \overline{\text{HTL}}_{sn, \geq 2000} - \overline{\text{HTL}}_{sn, \leq 500} \right) \quad (15)$$

841 To operationalize the empirical findings of Byrne, Parkinson, and Newall (1990)—who
 842 showed that listeners with high-frequency losses exceeding 70–95 dB HL require low-frequency
 843 speech cues—Open-NL incorporates a **Profound High-Frequency Bypass**:

$$PF_{bypass} = \max \left(0, \min \left(1, \frac{95 - \overline{\text{HTL}}_{sn, \geq 2000}}{25} \right) \right) \quad (16)$$

844 The low-frequency penalty is scaled over a 20 dB slope window, capped at 15 dB:

$$\text{LF}_{penalty} = \max \left(0, \min \left(1, \frac{\text{Slope} - 15}{20} \right) \right) \cdot 15 \cdot PF_{bypass} \quad (17)$$

845 The penalty is then tapered below 1000 Hz using $W_{LF}(f)$:

$$C''_{interp}(f) = C'_{interp}(f) - (\text{LF}_{penalty} \cdot W_{LF}(f)) \quad (18)$$

846 The resulting baseline target at 65 dB SPL is:

$$G_{65}^{(0)}(f) = 0.46 \cdot \text{HTL}_{sn}(f) + C''_{interp}(f) \quad (19)$$

847 Ablation of SD-LFP Constraints

848 Table S1 provides the reference audiometric profiles (A1–A7, Johnson & Dillon, 2011).
 849 Table S2 demonstrates the mechanical impact of the SD-LFP constraint on the initial heuristic
 850 seeds.

851 **TABLE S1. Reference Audiometric Profiles (Adapted from Johnson & Dillon,**
 852 **2011).** *Thresholds are in dB HL. Profiles A1–A5 are purely sensorineural. A6 is mixed (30*

853 *dB ABG). A7 is conductive (50 dB ABG).*

Profile	Type	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	ABG
A1	Flat	15	20	30	40	50	60	0 dB
	moder-							
	ate							
A2	Reverse	60	50	40	30	20	15	0 dB
	slope							
A3	Moderately	10	20	40	50	55	60	0 dB
	slop-							
	ing							
A4	Steeply	0	0	10	40	70	80	0 dB
	slop-							
	ing							
A5	Steeply	10	10	20	60	80	100	0 dB
	slop-							
	ing							
A6	Mixed	50	55	60	65	75	80	30 dB
A7	Conductive	50	50	50	50	50	50	50 dB

854 **TABLE S2. Ablation of SD-LFP Constraints on Optimizer Seed (65 dB SPL**
855 **Input).**

	Unconstrained	Unconstrained	SD-LFP Seed	SD-LFP Seed
Profile	Seed SII	Seed Sones	SII	Sones
A1 (Flat mod)	0.86	7.2	0.86	7.2
A2 (Reverse)	0.88	6.3	0.88	6.0
A3 (Mod sloping)	0.79	6.8	0.79	6.8
A4 (Steeply)	0.81	6.9	0.81	6.9
A5 (Steeply)	0.68	7.1	0.68	6.0
A6 (Mixed)	0.82	3.0	0.82	3.0
A7 (Conductive)	0.97	1.0	0.97	1.0

856

857 S.I.6. Stage 6: Severe-Loss Audibility Booster

858 To overcome inner hair cell loss in severe impairments, an opt-in, bounded **Severe-Loss**
859 **Booster** can be applied:

$$G_{slb}(f) = B_{en} \cdot 0.15 \cdot \max(0, \min(80, \text{HTL}_{sn}(f)) - T_{onset}) \quad (20)$$

860 where $B_{en} \in \{0, 1\}$ (default: 0, disabled), and $T_{onset} = 70$ dB HL (conservative default)
861 or 60 dB HL (aggressive ablation mode). The intermediate target is:

$$G_{65}^{(1)}(f) = G_{65}^{(0)}(f) + G_{slb}(f) \quad (21)$$

863 S.I.7. Stage 7: Soft-Compression High-Frequency Desensitization

864 To prevent unconstrained audibility maximization from prescribing intolerable high-
 865 frequency gain in steeply sloping losses, Open-NL applies a dynamic soft-compression envelope
 866 (L_{gain}):

$$L_{gain}(f) = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn}(f) - 60) \quad (22)$$

867 (*Note: For patients in “Moderate” or “High” distortion categories, L_{gain} is reduced by 10*
 868 *dB*).

869 Excess gain above this dynamic limit is:

$$\text{Excess}(f) = \max\left(0, G_{65}^{(1)}(f) - L_{gain}(f)\right) \quad (23)$$

870 The sloping factor $S_{factor}(f)$ relative to the best low-frequency threshold ($\text{HTL}_{bestlow} =$
 871 $\min_{f' \leq 1000} \text{HTL}_{sn}(f')$) and the high-frequency fade-in weight $W_{hf}(f)$ are:

$$S_{factor}(f) = \max\left(0, \min\left(1, \frac{\text{HTL}_{sn}(f) - \text{HTL}_{bestlow} - 25}{20}\right)\right) \quad (24)$$

$$W_{hf}(f) = \max\left(0, \min\left(1, \frac{f - 2000}{2000}\right)\right) \quad (25)$$

872 Applying 2:1 soft compression to the excess yields:

$$G_{65}^{(2)}(f) = G_{65}^{(1)}(f) - (W_{hf}(f) \cdot S_{factor}(f) \cdot 0.50 \cdot \text{Excess}(f)) \quad (26)$$

873 If explicit cochlear dead regions (f_{e_hf} , f_{e_lf}) or distortion categories (Margolis et al.,
874 2025) are defined, steep parametric roll-offs are applied:

$$G_{65}^{(2)}(f) \leftarrow G_{65}^{(2)}(f) - \max\left(0, \log_2\left(\frac{f}{1.7f_{e_hf}}\right)\right) \cdot 30 - \max\left(0, \log_2\left(\frac{0.57f_{e_lf}}{f}\right)\right) \cdot 30 \quad (27)$$

875

876 **S.I.8. Stage 8: Dynamic Range Mapping (LDL Squeeze)**

877 To accommodate reduced clinical dynamic ranges (DSL v5.0 philosophy; Scollie et al.,
878 2005), Open-NL predicts an HL-domain Loudness Discomfort Level:

$$\text{LDL}_{pred}(f) = 100 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 40) + \text{Loss}_{cond}(f) \quad (28)$$

879 When measured $\text{LDL}_{meas}(f)$ is lower than predicted, the dynamic range discrepancy
880 $\Delta_{LDL}(f)$ is quantified:

$$\Delta_{LDL}(f) = \text{LDL}_{meas}(f) - \text{LDL}_{pred}(f) \quad (29)$$

881 Target gain is attenuated by 0.2 dB per dB of dynamic range squeeze:

$$G_{65}^{(3)}(f) = G_{65}^{(2)}(f) + (0.2 \cdot \Delta_{LDL}(f)) \quad (30)$$

Simultaneously, the baseline compression ratio increases by +0.02 per dB of squeeze
(Stage 10).

S.I.9. Stage 9: Transducer Bandwidth Roll-off

Acoustic transducers physically struggle to reproduce frequencies at the extremes of the
spectrum (≤ 250 Hz and ≥ 6000 Hz), where massive gain leads to distortion and feedback.
Open-NL applies a continuous fractional bandwidth roll-off multiplier $M_{bw}(f)$ defined over
seven anchor frequencies:

$$\mathbf{f}_{bw} = [250, 500, 1000, 2000, 4000, 6000, 8000] \text{ Hz} \quad (31)$$

$$\mathbf{w}_{bw} = [0.7, 1.0, 1.0, 1.0, 1.0, 0.8, 0.5] \quad (32)$$

The closed-form continuous multiplier is obtained via piecewise log-linear interpolation:

$$M_{bw}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{bw}, \mathbf{w}_{bw}) \quad (33)$$

$$G_{65}^{(4)}(f) = G_{65}^{(3)}(f) \cdot M_{bw}(f) \quad (34)$$

S.I.10. Stage 10: Multi-Channel WDRC Mapping, Input/Output

Pivot, and Demographic Adjustments

Target gain at arbitrary overall input level L_{in} (e.g., 50, 65, 80 dB SPL) is computed using a multi-channel wide dynamic range compression architecture pivoted around the Long-Term Average Speech Spectrum (LTASS).

1. **Speech Spectrum Pivot:** Let $P(f)$ be the critical-band normal speech level at 65 dB SPL overall. The band-specific input level is:

$$L_{band}(f) = P(f) + (L_{in} - 65) \quad (35)$$

2. **Dynamic Compression Ratio Calculation:**

$$CR_{base}(f) = 1 + \frac{\max(0, HTL_{sn}(f) - 20)}{40} - (0.02 \cdot \Delta_{LDL}(f)) \quad (36)$$

For severe loss ($HTL_{sn} > 65$ dB HL), compression reduces back toward linear to preserve the temporal speech envelope (Keidser et al., 2007):

$$W_{freq}(f) = \max\left(0, \min\left(1, \frac{f - 500}{2500}\right)\right) \quad (37)$$

$$CR_{loud}(f) = CR_{base}(f) - \left(\frac{\max(0, HTL_{sn}(f) - 65)}{30} \cdot (1.5 - 0.5 \cdot W_{freq}(f))\right) \quad (38)$$

$CR_{loud}(f)$ is clamped between 1.0 and a maximum $CR_{max}(f) = 1.5 + 0.9 \cdot W_{freq}(f)$ (further relaxed for age > 60). In Comfort in Noise (CIN) mode, CR_{loud} is clamped to

906 ≤ 1.5 .

907 *(Note on Evidentiary Asymmetry: The 1.5:1 CIN clamp is an asserted engineering*
 908 *heuristic inspired by NAL-NL3 to curb listening fatigue in noise, lacking direct empirical*
 909 *derivation. In contrast, the global upper ceiling on compression ratios ($\leq 3.0 : 1$) is*
 910 *directly supported by extensive psychoacoustic literature [Souza, 2002; Souza et al.,*
 911 *2006], which demonstrates that compression ratios exceeding 3.0:1 cause severe temporal*
 912 *envelope flattening and acoustic contrast degradation).*

913 **3. Variable Compression Threshold (CT):** Overall CT ($CT_{overall}$) scales from 30
 914 to 45 dB SPL as a function of threshold. The band-level threshold is $CT_{band}(f) =$
 915 $P(f) + (CT_{overall} - 65)$ (reduced by 10 dB in CIN mode).

916 **4. Piecewise Linear-Compressive Spline:** Gain at the compression threshold is:

$$G_{CT}(f) = \begin{cases} G_{65}^{(4)}(f), & \text{if } CT_{band}(f) > P(f) \\ G_{65}^{(4)}(f) + (P(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)}\right), & \text{if } CT_{band}(f) \leq P(f) \end{cases} \quad (39)$$

917 The level-dependent WDRC target gain is:

$$G_{target}(f, L_{in}) = \begin{cases} G_{CT}(f), & \text{if } L_{band}(f) \leq CT_{band}(f) \\ G_{CT}(f) - (L_{band}(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)}\right), & \text{if } L_{band}(f) > CT_{band}(f) \end{cases} \quad (40)$$

5. Demographic Adjustments:

$$G_{target}(f, L_{in}) \leftarrow G_{target}(f, L_{in}) + \Delta_{gender} + \Delta_{config} + \Delta_{exp} \quad (41)$$

where $\Delta_{gender} = -1.5$ dB (female), $\Delta_{config} = +3.0$ dB (unilateral), and $\Delta_{exp} = -\min(6.0, 0.3 \cdot \max(0, \text{PTA}_{500,1k,2k} - 40))$ for new users.

TABLE S3. Effective Compression Ratios (50 to 80 dB SPL Inputs) across A1-A7 Audiograms. *Note: Dashes (-) indicate frequency regions where prescribed gain is exactly 0 dB for both 50 and 80 dB SPL inputs (linear amplification, CR = 1.0). Note that these values represent the emergent multi-level input/output ratios measured dynamically between 50 and 80 dB SPL inputs, rather than the prescribed static channel CRs calculated internally in Stage 10. To strictly enforce the 3.0:1 maximum bound, Open-NL applies dual constraints: a soft objective penalty (P_{cr}) during optimization, followed by a strict hard clamp, ensuring that pure intelligibility maximization never violates empirical psychoacoustic limits.*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-NL2	1.01	1.07	1.69	2.27	2.63	2.17
A1	Open-NL	1.25	-	1.19	1.54	2.27	4.35
A2	NAL-NL2	2.11	2.36	2.07	1.86	1.40	1.28

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A2	Open- NL	2.56	1.70	1.59	1.66	1.52	1.21
A3	NAL- NL2	-	1.10	1.88	2.48	2.70	2.19
A3	Open- NL	-	-	1.55	2.19	2.68	3.75
A4	NAL- NL2	-	-	1.12	2.17	2.22	1.88
A4	Open- NL	-	-	-	1.29	12.00	2.00
A5	NAL- NL2	-	-	1.59	2.11	1.99	1.79
A5	Open- NL	-	-	-	1.33	3.30	2.38
A6	NAL- NL2	1.32	1.42	1.61	2.00	2.24	1.91
A6	Open- NL	1.50	1.50	1.50	1.50	1.50	1.50
A7	NAL- NL2	1.00	1.00	1.00	1.00	1.00	1.00

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A7	Open-NL	1.00	1.00	1.00	1.00	1.00	1.00

S.I.11. Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits

Real-Ear Aided Responses (REAR) are heavily influenced by acoustic coupling. Low-frequency leakage is modeled by log-frequency interpolation over anchor frequencies $\mathbf{f}_{vent} = [250, 500, 1000, 2000, 4000, 8000]$ Hz:

$$V_{loss}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{vent}, \mathbf{v}_c) \tag{42}$$

where $\mathbf{v}_c$ is the coupling-specific attenuation vector defined in Table S4.

TABLE S4. Acoustic Coupling Real-Ear Insertion Loss Vectors ($\mathbf{v}_c$, in dB).

Coupling Configuration	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
custom_occluded	0	0	0	0	0	0
open_dome	-35	-28	-15	-2	0	0
tulip_dome	-25	-18	-5	0	0	0

Coupling						
Configu-						
ration	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
double_dome	-20	-10	0	0	0	0
vent_1mm_solid3		-1	0	0	0	0
vent_2mm_solid8		-2	0	0	0	0
vent_3mm_solid12		-4	0	0	0	0
vent_1mm_hollow12		-3	0	0	0	0
vent_2mm_hollow22		-12	-5	-2	0	0
vent_3mm_hollow25		-15	-8	-4	0	0

937 Conductive air-bone gaps are restored linearly with a 75% fraction: $G_{cond}(f) = 0.75 \cdot$
 938 $Loss_{cond}(f)$ (Note: This 75% restoration fraction is a pragmatic engineering convention—
 939 adapted from clinical practice to prevent excessive output demands and MPO clipping; Johnson,
 940 2013a—without direct empirical derivation from listener preference). To prevent active anti-
 941 phase cancellation demands and comb filtering, insertion gain is floored at $V_{loss}(f) - 10$
 942 dB:

$$G_{heuristic}(f, L_{in}) = \max(G_{target}(f, L_{in}) + 0.75 \cdot Loss_{cond}(f) + V_{loss}(f), V_{loss}(f) - 10) \quad (43)$$

943 Hardware receiver limits (MPO/SSPL90) are established to avoid severe saturation

944 distortion:

$$\text{MPO}(f) = \min(120, 105 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 20) + \text{Loss}_{cond}(f)) \quad (44)$$

945

946 **S.I.12. Stage 12: Embedded Nelder-Mead Simplex Optimization &** 947 **Physiological Loudness Ceilings**

948 When `optimize = TRUE`, Open-NL adjusts the heuristic targets by minimizing an uncon-
949 strained multi-objective loss function via Nelder-Mead simplex search (`stats::optim`).

950 **Parameter Vector and Gain Formation**

951 The optimization parameter vector is $\boldsymbol{\delta} = [\delta_{250}, \delta_{500}, \delta_{1000}, \delta_{2000}, \delta_{4000}, \delta_{8000}]^T \in \mathbb{R}^6$. During
952 evaluation, shifts are clamped:

$$\delta_{clamped,j} = \max(-60, \min(30, \delta_j)) \quad (45)$$

953 Candidate insertion gains $\mathbf{G} \in [0, 80]$ dB are interpolated to calculation frequencies:

$$G(f) = \max(0, \min(80, G_{heuristic}(f) + \delta_{clamped}(f))) \quad (46)$$

954 **Loss Function Formulation**

955 The Nelder-Mead solver minimizes:

$$\mathcal{L}(\boldsymbol{\delta}) = -100 \cdot \text{SII}_{desens}(\mathbf{G}) + P_{anchor} + P_{bounds} + P_{loud} + P_{spl} + P_{rough} + P_{order} + P_{cr} + P_{abg} \quad (47)$$

956 where SII_{desens} is the effective Speech Intelligibility Index calculated using the
 957 `johnson2011_smoothed` transfer function.

958 Explicit Penalty Terms and Exact Weights (λ)

959 1. Anchor Penalty ($\lambda_{anchor} = 0.1$):

$$P_{anchor} = 0.1 \sum_{j=1}^6 |\delta_j| \quad (48)$$

960 2. Out-of-Bounds Penalty ($\lambda_{bounds} = 1000.0$):

$$P_{bounds} = 1000.0 \left[\sum_{j=1}^6 \max(0, \delta_j - 30)^2 + \sum_{j=1}^6 \max(0, -\delta_j - 60)^2 \right] \quad (49)$$

961 3. Physiological Loudness Ceiling Penalty ($\lambda_{loud} = 2000.0$):

$$P_{loud} = 2000.0 \cdot \max(0, \text{Sones}_{MG04}(\mathbf{G}) - \text{Cap}) \quad (50)$$

962 *(Note: While gain-based penalties in this framework are squared to strongly penalize*
 963 *large deviations, the loudness penalty is explicitly linear. This is a deliberate choice:*
 964 *because sones inherently represent a compressive power-law transformation of physical*
 965 *acoustic energy, a linear penalty in the sone domain naturally exerts an exponentially*

growing restriction on the underlying insertion gain. Squaring the sone error introduces severe mathematical stiffness and destabilizes the simplex gradient.) where Sones_{MG04} is the Moore & Glasberg (2004) specific-loudness integration computed via native C++. The dynamic U-shaped cap is interpolated across Pure Tone Average knots $\mathbf{PTA}_{knots} = [10, 32.5, 52.5, 72.5, 90]$ dB HL from level-specific sone vectors:

- $L_{in} = 50$ dB SPL: $\mathbf{K}_{50} = [1.5, 1.0, 0.8, 1.2, 1.2]$ sones
- $L_{in} = 65$ dB SPL: $\mathbf{K}_{65} = [7.0, 4.5, 4.0, 6.5, 6.0]$ sones
- $L_{in} = 80$ dB SPL: $\mathbf{K}_{80} = [20.0, 12.0, 10.0, 15.0, 14.0]$ sones

$$\text{Cap}_{base} = \text{interp}(\text{PTA}_{sn}(\text{defined as the four - frequency pure - tone average of the sensorineural hearing loss}), \mathbf{K}) \quad (51)$$

Profile Adjustments:

- Reverse-slope restriction ($L_{in} \geq 75$ dB SPL and low-to-high threshold difference > 10 dB): $\text{Cap} = \text{Cap}_{base} - 0.10 \cdot (\overline{\text{HTL}}_{\leq 500} - \overline{\text{HTL}}_{\geq 4000})$.
- Conductive air-bone gap adjustment: $\text{Cap} \leftarrow \text{Cap} - 0.25 \cdot \text{PTA}_{ABG}$ (if $L_{in} \geq 75$ dB SPL) or $+0.10 \cdot \text{PTA}_{ABG}$ (if $L_{in} < 75$ dB SPL).

4. Broadband SPL Saturation Ceiling Penalty ($\lambda_{spl} = 2000.0$):

$$P_{spl} = 2000.0 \cdot \max(0, \text{SPL}_{aided} - 110.0) \quad (52)$$

5. **Spectral Roughness Penalty** ($\lambda_{rough} = 0.5$):

$$P_{rough} = 0.5 \sum_{j=1}^5 (\delta_{j+1} - \delta_j)^2 \quad (53)$$

6. **Inter-Level Order Monotonicity Penalty** ($\lambda_{order} = 2000.0$): Enforces $G_{50} \geq G_{65}$

and $G_{80} \leq G_{65}$:

$$P_{order} = \begin{cases} 2000.0 \sum_{j=1}^6 \max(0, G_{65,j} - G_{50,j})^2, & \text{for } L_{in} = 50 \\ 2000.0 \sum_{j=1}^6 \max(0, G_{80,j} - G_{65,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (54)$$

7. **Compression Ratio Ceiling Penalty** ($\lambda_{cr} = 200.0$):

$$P_{cr} = \begin{cases} 200.0 \sum_{j=1}^6 \max(0, (G_{50,j} - G_{65,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 50 \\ 200.0 \sum_{j=1}^6 \max(0, (G_{65,j} - G_{80,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (55)$$

where $\Delta_{max,j} = 10.0 \cdot (1 - \text{ABG}_j / \max(0.001, \text{HTL}_j))$ for soft speech, bounding emergent compression ratios safely below 3.0:1 for sensorineural loss while enforcing linear amplification for conductive components. This 3.0:1 ceiling represents the best-supported parameter in the algorithm, firmly grounded in empirical psychoacoustic literature (Souza, 2002; Souza et al., 2006) demonstrating severe envelope flattening, loss of acoustic contrast, and speech-in-noise deficits for CRs $> 3.0 : 1$.

8. **Air-Bone Gap Excursion Penalty** ($\lambda_{abg} = 1.0$):

$$P_{abg} = \begin{cases} 1.0 \sum_{j=1}^6 \max(0, \delta_j)^2, & \text{if } \exists j, \text{Loss}_j > 0 \\ 0, & \text{otherwise} \end{cases} \quad (56)$$

Solver Controls and Convergence Tolerances

- **Algorithm:** Nelder-Mead Simplex via R's `stats::optim()`.
- **Iteration Ceiling:** `control = list(maxit = 800)`.
- **Convergence Tolerance:** Relative convergence tolerance `reltol = sqrt(.Machine$double.eps)` $\approx 1.49 \times 10^{-8}$.
- **Initial Simplex Seeding:** Seeding incorporates an audibility projection $\delta_{start} = \min(20, \max(0, \text{target_aided} - G_{heuristic}))$, where $\text{target_aided} = \min(\text{UCL} - 5, \max(L_{in}, \text{HTL} + 10))$. Soft inputs receive +3 dB shift; loud inputs receive $\max(-10, \delta_{start} - 5)$ dB shift.
- **Multi-Start Strategy:** A 5-iteration multi-start routine (seeding the initial simplex with the NAL-R target and executing four additional randomized restarts with uniform random jitter $\delta_{start} \leftarrow \delta_{start} + \mathcal{U}(-5, +5)$) is deployed to stabilize numerical convergence, though Nelder-Mead inherently lacks formal global convergence guarantees.

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